

Ryan Quinn

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Education

Massachusetts Institute of Technology (MIT)

Graduating May 2027

Bachelor's in Computer Science, Artificial Intelligence & Machine Learning

- **Relevant Coursework:** Computer Vision (6.8300 & 6.S058), Computer Graphics (6.4400), Robotic Manipulation (6.4210), Machine Learning (6.3900), Probability and Inference (6.3700), Software Development (6.1020), Linear Algebra (18.06), Algorithms (6.1210), Philosophy of Quantum Mechanics (24.111)
- **Extracurricular:** MIT Entrepreneurship Club; Chi Phi Fraternity (House Manager); Varsity Lacrosse (2 years)

Technical Experience

MIT Computer Science & Artificial Intelligence Lab (CSAIL)

January 2025 – February 2026

Machine Learning Undergraduate Researcher

- Engineered a custom differentiable simulator that optimizes physics for 3D generative diffusion models.
- Designed algorithms for adaptive octree structures that refine around complex object boundaries.
- Built flowSim (C/Python) to generate large-scale training data for m3, a machine learning model for computing flow completion times; produced millions of labeled flows for experiments.
- Added multi-class traffic support to create more realistic training scenarios.

Amazon

May 2025 – August 2025

Software Development Engineer Intern

- Launched two new mission strategies for desktop homepage recommendations (~1000 impressions/sec), contributing to \$1M+ in company revenue.
- Reduced feature-store lookup latency by 86% (35 to 5 ms), improving homepage recommendation responsiveness and cutting cloud spend by ~\$50,000/month.
- Refactored Java backend services to simplify integrations and unblock new feature rollouts.

Novelis

June 2024 – February 2025

Software Engineer Intern

- Developed a C# RESTful API to automate resource assignment for ingot production; streamlined production planning workflows.
- Integrated SQL (CRUD) and Dapper to implement business logic for optimizing ingot chemistries.
- Diagnosed and resolved MES data-handling issues at a new \$400M manufacturing plant to stabilize production.

Technical Projects

VisionBeam (Autonomous Lighting System)

June 2025 – October 2025

- Engineered a deep learning tracking system enabling DMX hardware to autonomously follow human targets in dynamic, high-occlusion environments.
- Integrated YOLOv8 and ByteTrack for real-time object detection and multi-object tracking, ensuring consistency in crowded environments.
- Designed an activity-scoring heuristic (combining bounding boxes with frame differencing) to generate spatial heatmaps of peak human motion.
- Developed a calibration and inverse kinematics engine (OpenCV, SciPy) mapping 2D camera pixels to 3D world coordinates, allowing DMX hardware to operate at 40 FPS.

Technical Skills

Languages: Python, TypeScript, SQL, C++, C, Java, JavaScript, C#, HTML, CSS, Go

Tools: PyTorch, OpenCV, TensorFlow, NumPy, Docker, AWS, Flask, Linux, MySQL, DynamoDB, React, Node.js, Next.js, Swagger API, Dapper, Blender, Unity